Time Series HW7

Matvei Fedortsov

March 15, 2026

1 Task 1

We are given the estimated reduced-form VAR(1) model for two variables:

$$\begin{bmatrix} y_{t,1} \\ y_{t,2} \end{bmatrix} = \begin{bmatrix} 0.2 & 0.4 \\ 0.2 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1,1} \\ y_{t-1,2} \end{bmatrix} + \begin{bmatrix} \varepsilon_{t,1} \\ \varepsilon_{t,2} \end{bmatrix}, \qquad C_\varepsilon = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 4.25 \end{bmatrix}.$$

The second variable enters the first equation without a lag, but the first does not affect the second contemporaneously.

A recursive SVAR(1) can be written as

$$By_t = A_1 y_{t-1} + u_t, \quad u_t \sim (0, D), \; D = \mathrm{diag}(d_1, d_2),$$

with

$$L = \begin{bmatrix} 1 & -a \\ 0 & 1 \end{bmatrix}, \quad L^{-1} = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}.$$

The reduced-form errors are $\varepsilon_t = B^{-1} u_t$, hence

$$C_\varepsilon = L^{-1} G (L^{-1})^\top.$$

$$L = \begin{bmatrix} 1 & 2/17 \\ 0 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 16/17 & 0 \\ 0 & 17/4 \end{bmatrix}.$$

SVAR(1):

$$\begin{cases} y_{t,1} = \dfrac{2}{17}\, y_{t,2} + \dfrac{19}{85}\, y_{t-1,1} + 0.4\, y_{t-1,2} + u_{t,1}, \\ y_{t,2} = 0.2\, y_{t-1,1} + u_{t,2}, \end{cases}$$

with structural shocks having covariance matrix $\mathrm{diag}(16/17,\ 17/4)$.

2 Task 2

2.1 Data

- Gold – gold price index,
- USTBarclaysTR – U.S. Treasury Bond Total Return Index,
- DJIA – Dow Jones Industrial Average.

First differences are denoted as d1_Gold, d1_USTBarclaysTR, d1_DJIA.

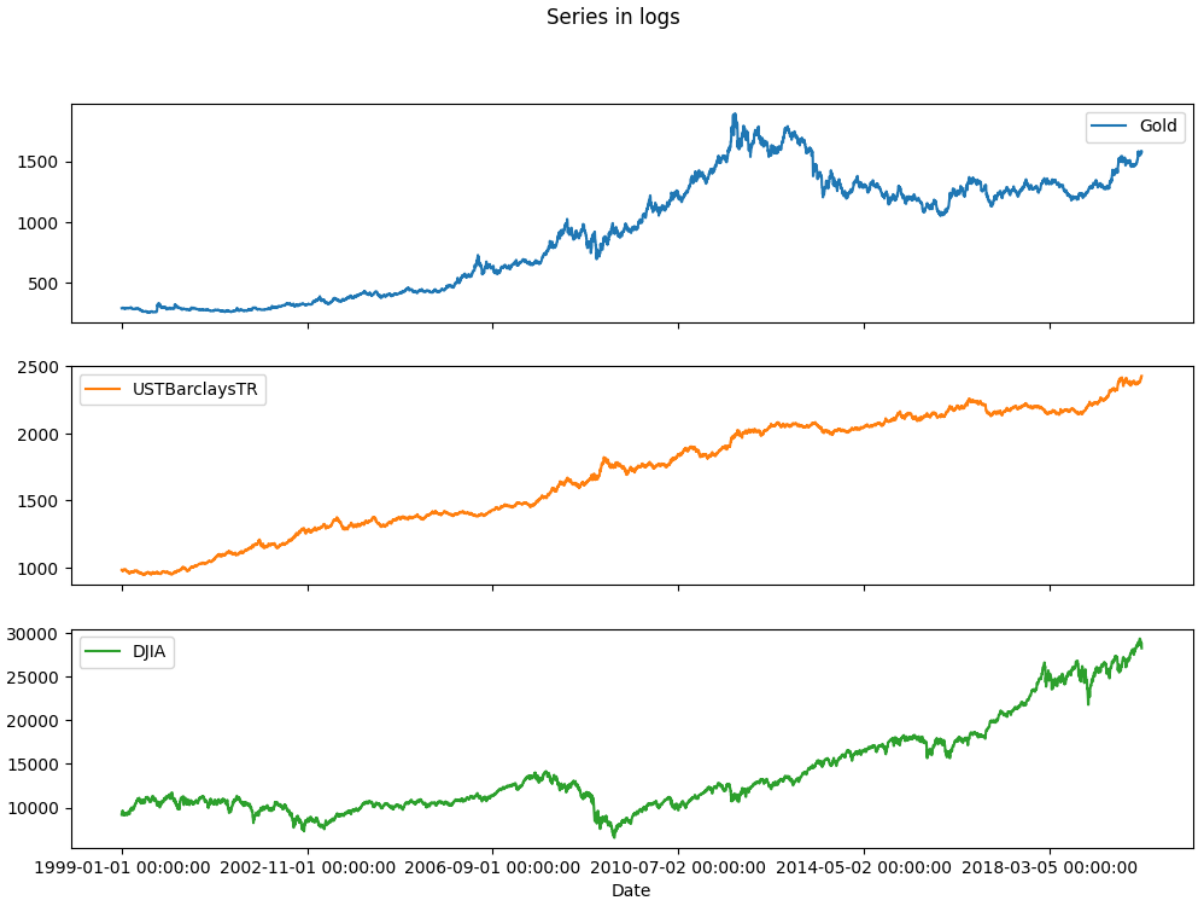

Figure 1: Data

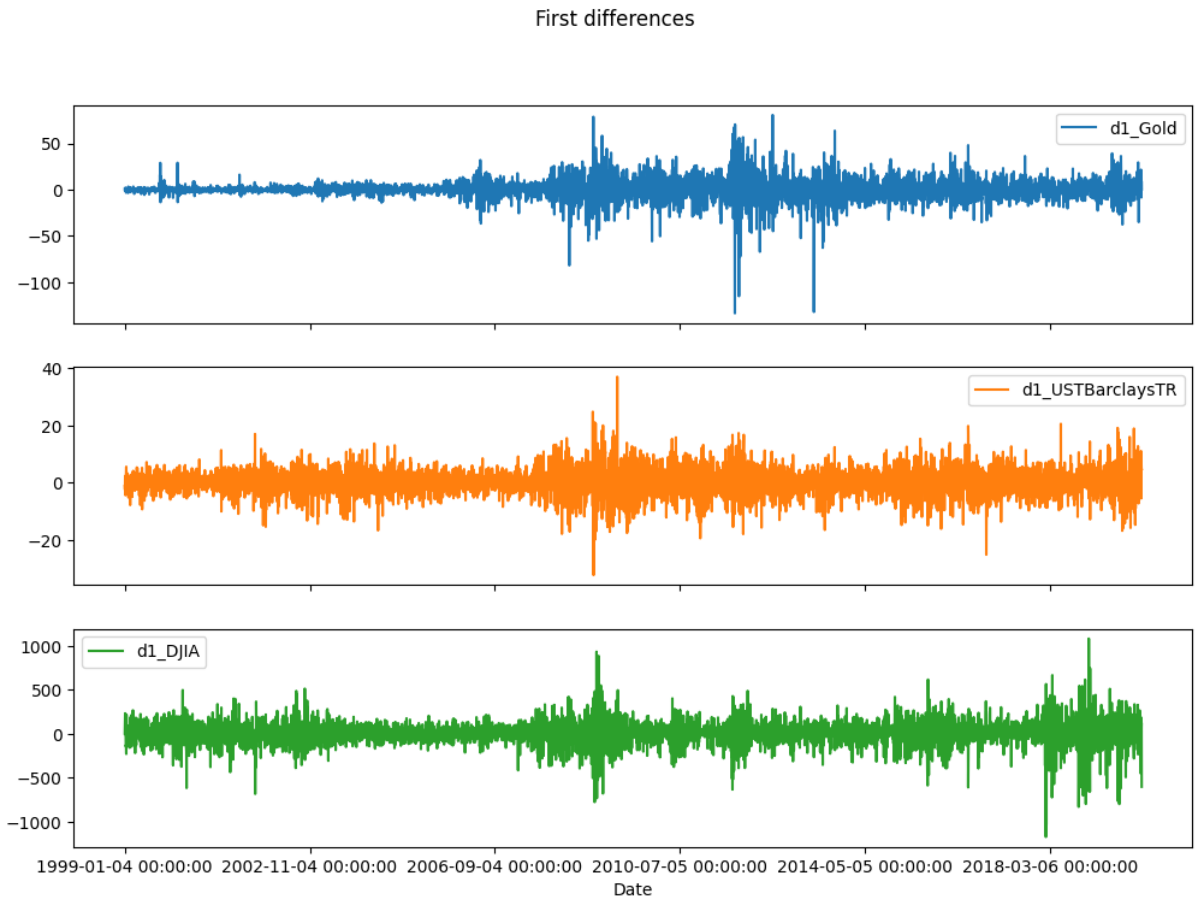

Figure 2: First differences

2.2 Stationarity Testing

Augmented Dickey–Fuller (ADF) tests were performed on both levels and first differences. The results are summarised in Table 1.

Table 1: ADF test p-values

Variable	Levels (p-value)	First differences (p-value)
Gold	0.855	0.000
USTBarclaysTR	0.923	0.000
DJIA	0.996	0.000

All series are non-stationary in levels but become stationary after one differencing, i.e. they are integrated of order one, I(1).

2.3 Choice of Influencing Variable

We are modelling d1_Gold. CCF and Granger causality tests were performed to choose which variable explains better. Table 2 reports the p-values for lags 1–12.

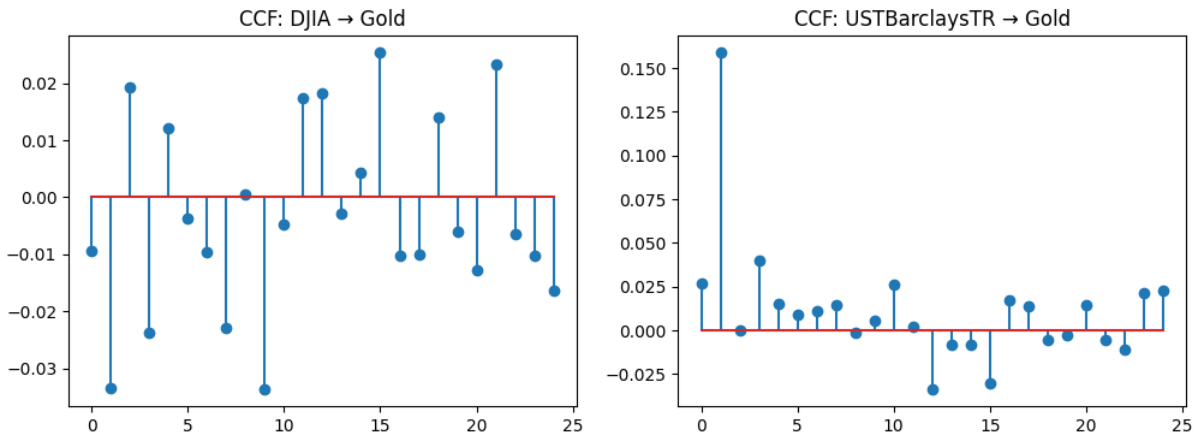

Figure 3: CCF

Table 2: Granger causality test p-values (predictor → Gold)

Lag	1	2	3	4	5	6	7	8	9	12
DJIA → Gold	0.102	0.066	0.093	0.167	**0.047**	**0.011**	**0.009**	**0.015**	**0.027**	**0.031**
USTBarclaysTR → Gold	0.207	0.135	0.200	**0.007**	**0.008**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

The p-values for USTBarclaysTR become extremely small (below 0.01) from lag 4 onward, indicating a very strong and consistent causal relationship. Therefore, USTBarclaysTR was selected as the primary predictor.

2.4 Bivariate VAR Model (d1_Gold, d1_USTBarclaysTR)

The optimal lag length was chosen using information criteria (Table 3). The Akaike Information Criterion (AIC) reaches its minimum at lag 11, while BIC and HQIC suggest lag 1. Following the common practice of minimising AIC for forecasting purposes, we selected $p = 11$.

Table 3: Lag order selection for bivariate VAR (Gold, USTBarclaysTR)

Lag	AIC	BIC	HQIC	FPE
0	7.862	7.864	7.863	2596
1	7.835	7.842	7.838	2528
2	7.835	7.847	7.839	2527
...	...	...	...	...
11	**7.829**	7.885	7.849	**2513**
12	7.830	7.890	7.851	2514

The estimated VAR(11) coefficients (see Table **??**) indicate that many lags of USTBarclaysTR are statistically significant in the equation for d1_Gold, confirming its predictive power. The residual correlation between equations is very low (0.03).

Table 4: Estimated coefficients for the d1_Gold equation in VAR(11)

Variable	Lag	Coefficient	Std. Error	t-statistic	p-value
const	–	0.058689	0.148249	0.396	0.692
d1_Gold	1	-0.031837	0.013524	-2.354	0.019
d1_UST	1	0.379166	0.031540	12.022	0.000
d1_Gold	2	-0.026109	0.013527	-1.930	0.054
d1_UST	2	0.032905	0.031966	1.029	0.303
d1_Gold	3	0.020768	0.013530	1.535	0.125
d1_UST	3	0.124352	0.031977	3.889	0.000
d1_Gold	4	-0.034031	0.013533	-2.515	0.012
d1_UST	4	0.033991	0.032004	1.062	0.288
d1_Gold	5	0.003392	0.013554	0.250	0.802
d1_UST	5	0.033918	0.031997	1.060	0.289
d1_Gold	6	-0.024186	0.013551	-1.785	0.074
d1_UST	6	0.038997	0.031982	1.219	0.223
d1_Gold	7	-0.001751	0.013568	-0.129	0.897
d1_UST	7	0.057029	0.031972	1.784	0.074
d1_Gold	8	0.008021	0.013570	0.591	0.554
d1_UST	8	0.004073	0.031958	0.127	0.899
d1_Gold	9	0.025600	0.013559	1.888	0.059
d1_UST	9	0.016758	0.031971	0.524	0.600
d1_Gold	10	-0.019322	0.013562	-1.425	0.154
d1_UST	10	0.053783	0.031938	1.684	0.092
d1_Gold	11	-0.038495	0.013381	-2.877	0.004
d1_UST	11	0.009939	0.031930	0.311	0.756

Table 5: Estimated coefficients for the d1_USTBarclaysTR equation in VAR(11)

Variable	Lag	Coefficient	Std. Error	t-statistic	p-value
const	–	0.273426	0.063610	4.298	0.000
d1_Gold	1	-0.005668	0.005803	-0.977	0.329
d1_UST	1	-0.031423	0.013533	-2.322	0.020
d1_Gold	2	-0.010755	0.005804	-1.853	0.064
d1_UST	2	-0.022650	0.013716	-1.651	0.099
d1_Gold	3	0.001156	0.005805	0.199	0.842
d1_UST	3	0.015460	0.013720	1.127	0.260
d1_Gold	4	0.018896	0.005806	3.254	0.001
d1_UST	4	0.006056	0.013732	0.441	0.659
d1_Gold	5	0.001733	0.005816	0.298	0.766
d1_UST	5	-0.029465	0.013729	-2.146	0.032
d1_Gold	6	-0.021158	0.005814	-3.639	0.000
d1_UST	6	-0.006072	0.013723	-0.443	0.658
d1_Gold	7	0.016903	0.005822	2.903	0.004
d1_UST	7	0.014059	0.013718	1.025	0.305
d1_Gold	8	0.015210	0.005823	2.612	0.009
d1_UST	8	-0.014298	0.013712	-1.043	0.297
d1_Gold	9	0.001910	0.005818	0.328	0.743

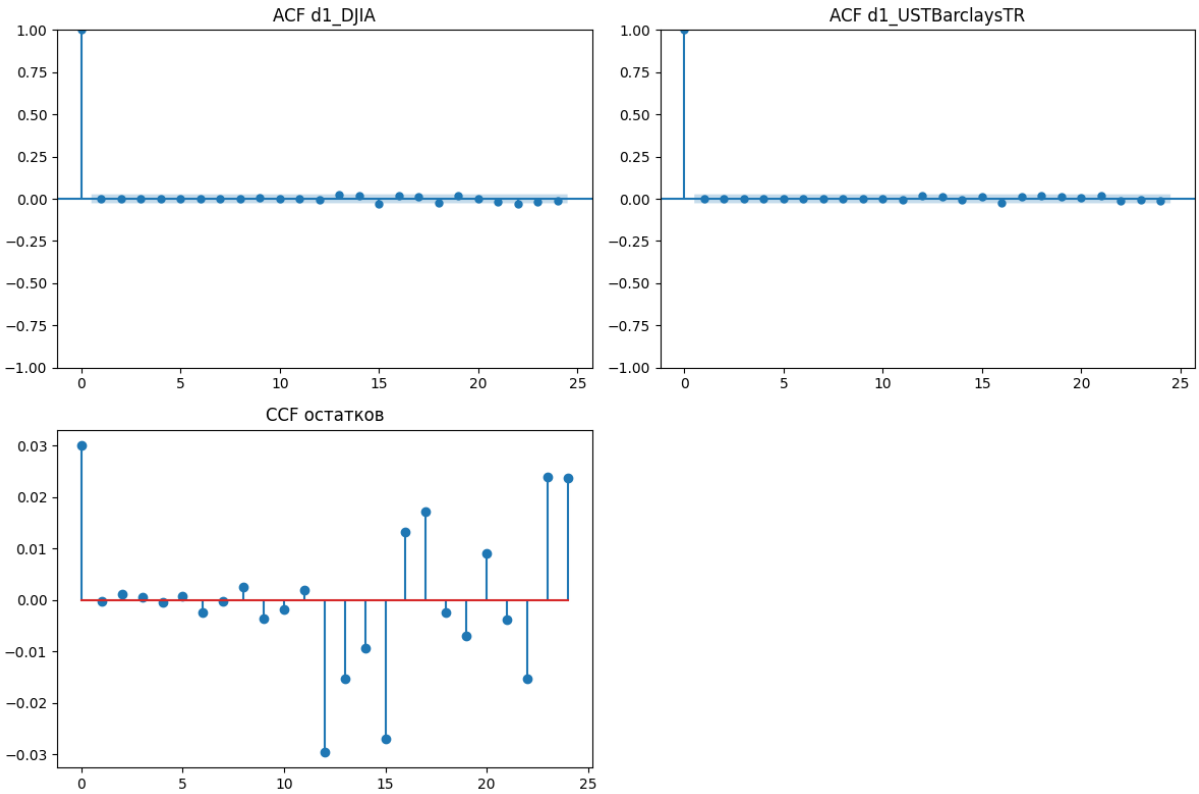

Figure 4: Residuals

Table 5 – continued from previous page

Variable	Lag	Coefficient	Std. Error	t-statistic	p-value
d1_UST	9	-0.005636	0.013718	-0.411	0.681
d1_Gold	10	-0.001522	0.005819	-0.262	0.794
d1_UST	10	0.026668	0.013704	1.946	0.052
d1_Gold	11	-0.008736	0.005741	-1.522	0.128
d1_UST	11	0.001765	0.013700	0.129	0.897

2.5 Residual Diagnostics

To check for remaining autocorrelation and cross-correlation, the multivariate portmanteau tests of Hosking and Li–McLeod were applied using $2p = 22$ lags. The results are:

$$Q_{\text{Hosking}} = 1.4969 \times 10^{12},\ p = 0.000; \qquad Q_{\text{LiMcLeod}} = 1.4969 \times 10^{12},\ p = 0.000.$$

The null hypothesis of no autocorrelation is strongly rejected, indicating that the model does not fully capture the dynamics. In a real-world study one would either increase the lag order further or consider other specifications (e.g., VECM if cointegration is present). For the purpose of this assignment we proceed with the current specification.

2.6 Inclusion of the Third Variable (DJIA)

A trivariate VAR including d1_DJIA was estimated. The AIC selected lag 6 for this model. To justify whether adding DJIA improves the model, an F-test comparing the restricted (bivariate) and unrestricted (trivariate) models was performed on the common sample. The outcome was

$$F = -905.6, \quad p = 1.000.$$

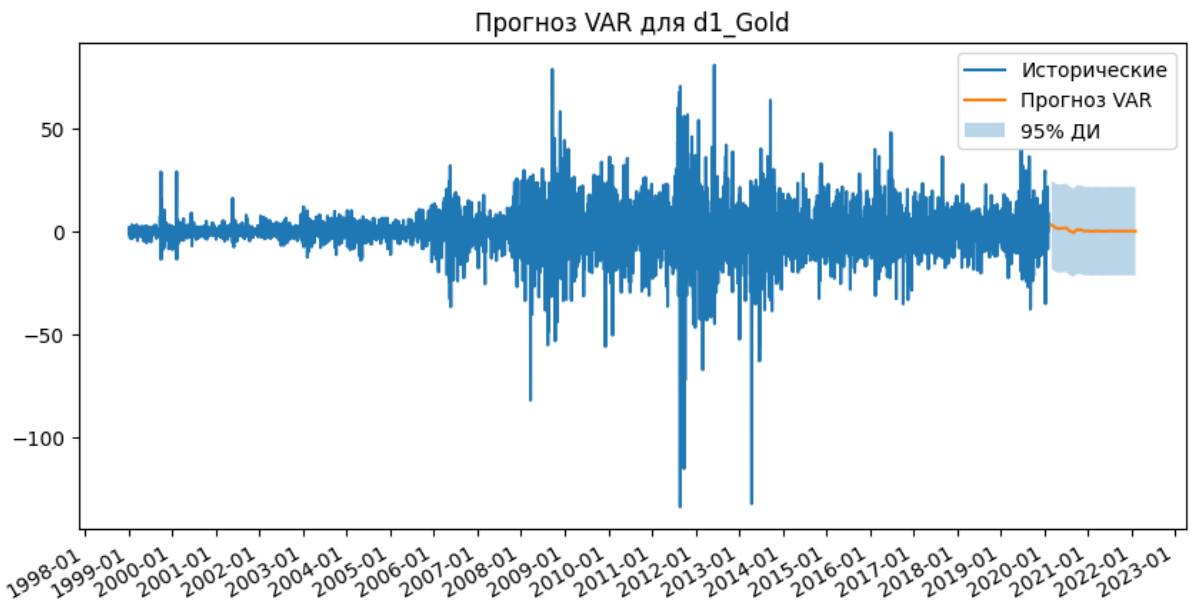

Figure 5: Diff forecast

The negative F-statistic and p-value of 1 indicate that the inclusion of DJIA does not lead to a statistically significant improvement. Hence the bivariate VAR(11) is retained as the final model.

2.7 Forecasting

Using the selected VAR(11) model, an interval forecast for the next 24 months was generated for d1_Gold. The forecast for the level of Gold was obtained by cumulatively summing the predicted differences starting from the last historical value. The forecasts (with 95% confidence intervals) are shown in Figure **??**.

The confidence bands widen with the forecast horizon, reflecting the accumulation of uncertainty. The forecast for the level suggests a gradual increase over the next two years, although the wide intervals indicate substantial uncertainty.

2.8 Impulse Response Functions and Variance Decomposition

Structural interpretation is based on orthogonalised impulse responses (OIRF) and forecast error variance decomposition (FEVD). A recursive identification scheme was adopted with the ordering: d1_USTBarclaysTR → d1_Gold. This ordering reflects the assumption that the bond market (USTBarclaysTR) is more exogenous and does not react contemporaneously to shocks in the gold market, whereas gold may respond instantly to bond market shocks.

Figure **??** shows the effect of a one-standard-deviation shock to d1_USTBarclaysTR on d1_Gold. The response is positive and decays after about 4–5 months.

The FEVD (Table 6) reveals that own shocks explain about 97% of the forecast error variance of Gold at all horizons; the contribution of USTBarclaysTR remains around 3%, increasing slightly over time.

Figure 6: Gold price forecast

Table 6: FEVD for d1_Gold (selected horizons)

Horizon	Own contribution	Contribution of USTBarclaysTR
1	0.999	0.001
4	0.972	0.028
8	0.971	0.029
12	0.971	0.029

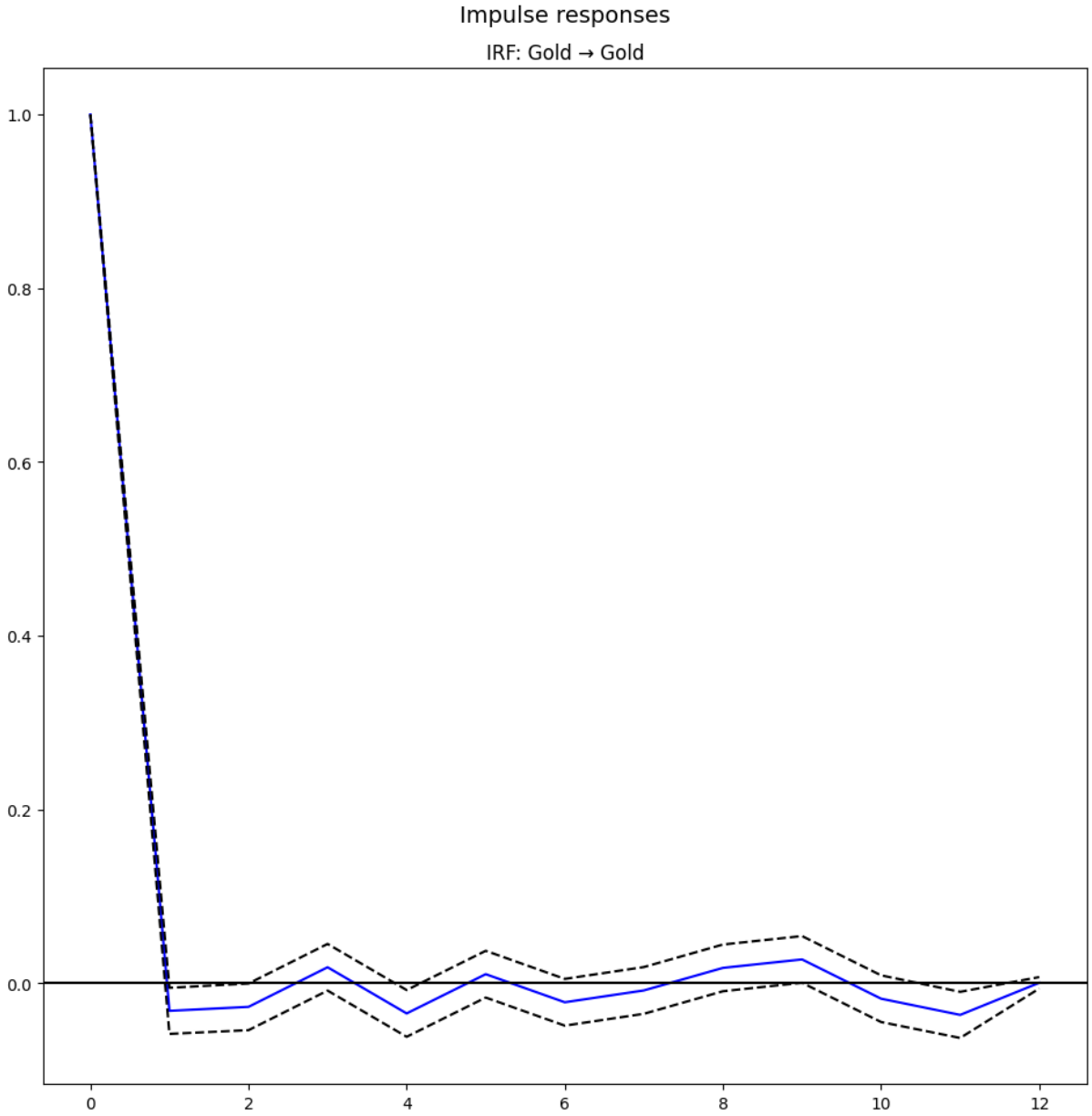

Figure 7: Enter Caption

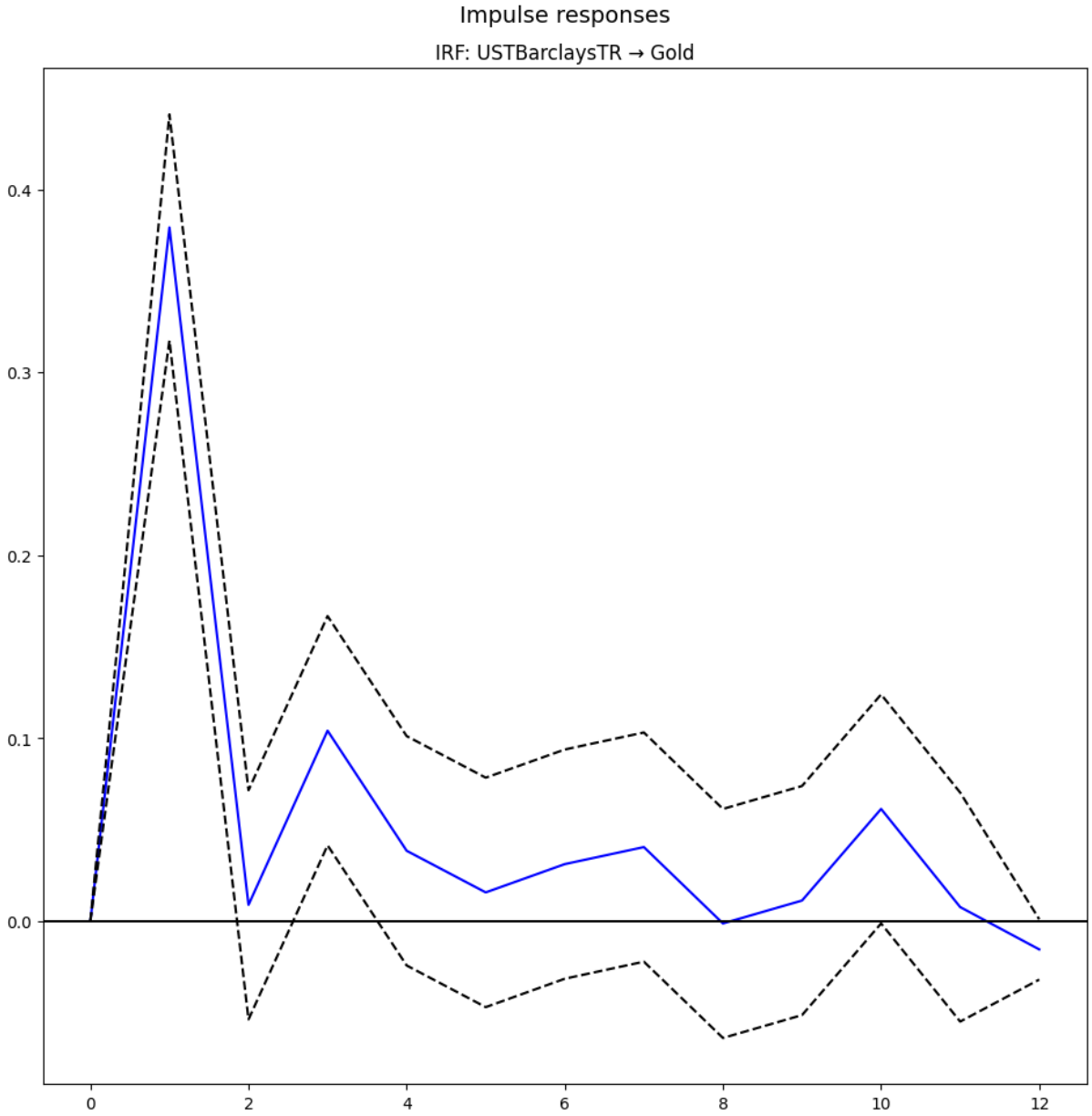

Figure 8: Enter Caption

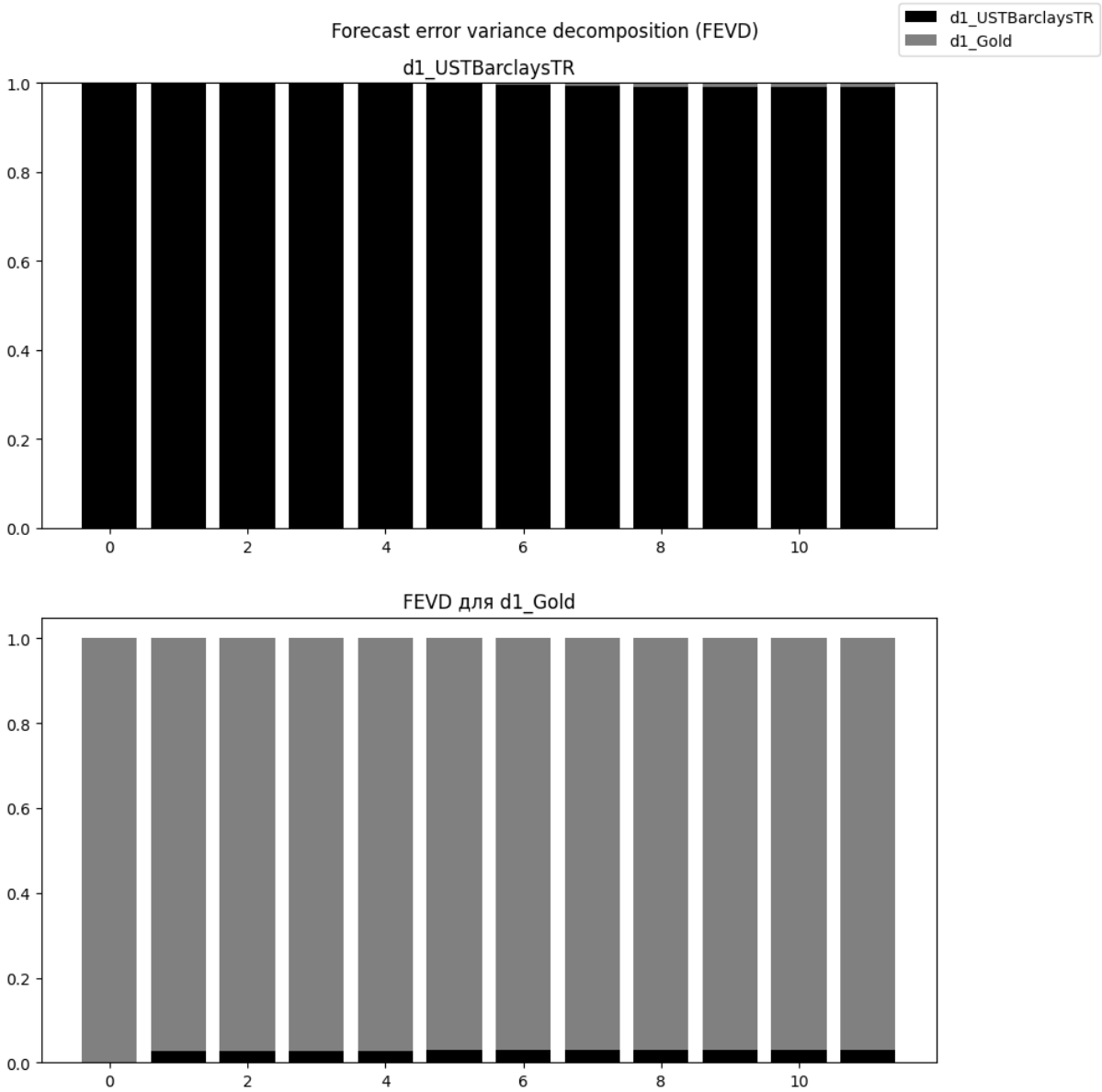

Figure 9: Enter Caption